Experiment 6

Workflow Automation Layer for the Vehicle Service Request Application

Aim

To design and build a complete automation layer in Zoho Creator for the Vehicle Service Request application, demonstrating all eight Workflow options: Form Workflows, Schedules, Approvals, Payments, Blueprints, Batch Workflows, Report Workflows and Functions.

Procedure

1. **On_ServiceRequest_Submit_Workflow (Form Workflow):** fires when a record is added; emails the owner their Service Request ID and next steps, and notifies the front office.
2. **Daily_Service_Reminder_Schedule (Schedule):** runs every evening and reminds owners with a service booking the next day, with drop-off time, bay details and a document checklist.
3. **Service_Estimate_Approval (Approval):** only requests whose Net Service Cost Payable is above the threshold need Service Manager approval before work begins.
4. **Service_Fee_Payment_Workflow (Payments):** generates a payment link from Net_Service_Cost_Payable, updates Payment_Status on success and flags failures for follow-up.
5. **Service_Lifecycle_Blueprint (Blueprint):** restricts each stage transition (Requested, In Diagnosis, In Repair, Ready for Pickup, Delivered) to specific roles and gives an auditable trail.
6. **Bulk_ReadyForPickup_Batch (Batch Workflow):** notifies owners of all Ready for Pickup vehicles in batches to avoid timeouts.
7. **OnEdit_Recalculate_NetCost_Workflow (Report Workflow):** keeps Net_Service_Cost_Payable accurate when the discount percentage is edited in the report grid.
8. **Generate_ServiceInvoice_Function (Function):** a reusable Deluge function called on status change to Delivered, from a resend button, or from the batch workflow.

Workflow Summary

Workflow Type	Name	Trigger	Action
Form Workflow	On_ServiceRequest_Submit_Workflow	On record add	Send confirmation mail with Service Request ID; notify front office
Schedule	Daily_Service_Reminder_Schedule	Every day, 6:00 PM	Remind owners with next-day bookings
Approval	Service_Estimate_Approval	Net cost above threshold	Manager approves or rejects; mail to owner
Payments	Service_Fee_Payment_Workflow	On payment request	Payment link; update Payment_Status
Blueprint	Service_Lifecycle_Blueprint	Status transitions	Role-restricted stages with audit trail
Batch Workflow	Bulk_ReadyForPickup_Batch	Scheduled / on demand	Batch notification to ready owners
Report Workflow	OnEdit_Recalculate_NetCost_Workflow	On edit in report	Recompute Net_Service_Cost_Payable
Function	Generate_ServiceInvoice_Function	Called by other workflows	Generate and mail service invoice

Sample Deluge: Form Workflow (On Success)

```
sendmail
[
    from : zoho.adminuserid
    to : input.Email_Address
    subject : "Service Request Received - " + input.Service_Request_ID
    message : "Dear " + input.Owner_Name.first_name + ", your request for vehicle "
              + input.Vehicle_Registration_Number + " is registered. "
              + "Our team will share the estimate before any work begins."
]
```

Process Diagram

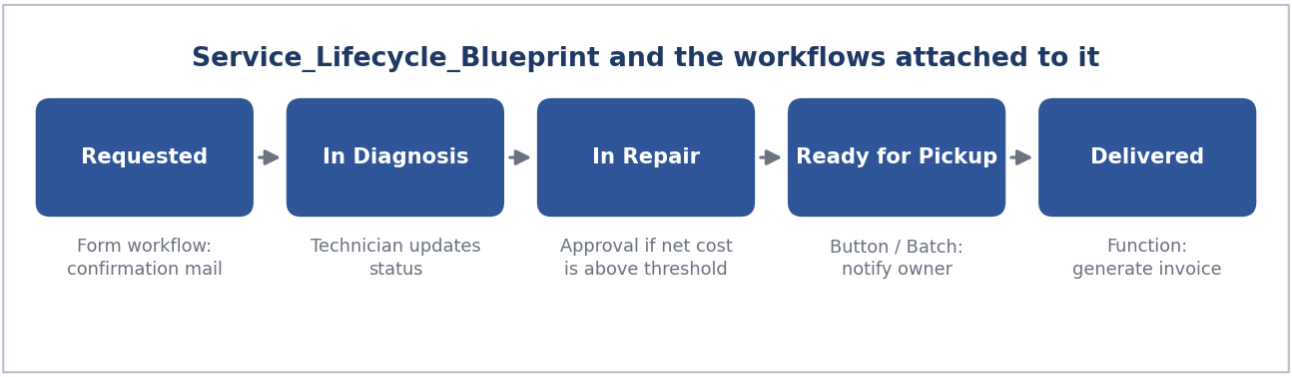


Figure 1: Diagram of Service_Lifecycle_Blueprint and the workflows attached to it (illustrative diagram, not a screenshot).

Workflow Tests and Expected Outcome

Workflow	Test Action	Expected Outcome
On_ServiceRequest_Submit_Workflow	Submit a new request	Owner receives a mail with the Service Request ID; front office is notified
Daily_Service_Reminder_Schedule	Booking dated tomorrow exists	Reminder with drop-off time and document checklist is sent in the evening
Service_Estimate_Approval	Net cost above the threshold	Request goes to the Service Manager; owner is mailed on approve or reject
Service_Fee_Payment_Workflow	Payment requested	Payment link generated; Payment_Status updated on success (needs a paid plan)
Service_Lifecycle_Blueprint	Move a request to the next stage	Only the permitted role can make the transition; trail is recorded
Bulk_ReadyForPickup_Batch	Run for all Ready for Pickup records	Owners notified in batches without timeout
OnEdit_Recalculate_NetCost_Workflow	Edit discount % in the report grid	Net_Service_Cost_Payable is recalculated
Generate_ServiceInvoice_Function	Status changed to Delivered	Invoice is generated and mailed to the owner

Blueprint Transitions and Roles

From Stage	To Stage	Allowed Role
Requested	In Diagnosis	Technician
In Diagnosis	In Repair	Technician (after estimate approval)
In Repair	Ready for Pickup	Technician
Ready for Pickup	Delivered	Front Office

Reminder Email (Expected)

```
Subject: Service Reminder - Tomorrow

Dear Priya Sharma,
Your vehicle TN 38 CD 1234 is booked for service tomorrow.
Please drop off the vehicle at the scheduled time and carry
the RC book and insurance copy.

Regards,
Vehicle Service Center
```

Result

A complete automation layer covering all eight Workflow options was created for the Vehicle Service Request application in Zoho Creator.